

Effect of Hair and Pen Mark Removal on Skin Lesion Classification

Group: Dinosaurs

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1 Introduction

When a dermatologist looks at a suspicious skin lesion, they mentally run through a checklist: is it asymmetric, does the border look irregular, are there multiple colors, how big is it? This is the ABCD rule and it has been used in clinical practice for decades (Nachbar et al., 1994). In this project, we try to automate part of that process by building a classifier that takes an image of a lesion and predicts whether it is cancerous or not.

The dataset we work with, PAD-UFES-20 (Pacheco et al., 2020), comes from a real clinical setting in Brazil. The images were taken on smartphones rather than specialist dermoscopy equipment, which means they are messier. One thing we noticed immediately when going through the data is how often hair and pen marks appear in the images. Pen marks are drawn by clinicians to outline lesions before procedures. Hair naturally grows on the skin and can partially cover the lesion area. Neither has anything to do with whether it is malignant, but both can interfere with the features we extract: hair creates false edges that interfere with asymmetry and border measurements, and pen marks land inside the lesion area and shift the colour statistics. A model that learns to use these patterns instead of the actual lesion is likely to perform worse on images from a different clinic where these artefacts look different (Winkler et al., 2019).

This is what led us to our research question:

“How much does the removal of hair and pen marks prior to feature extraction improve model performance?”

To answer it, we build a **baseline model** using features extracted directly from the original images and an **extended model** using the same features but extracted after cleaning the images first. The extended model also includes hair and pen mark coverage values as extra inputs.

One thing that distinguishes our approach is that we treat hair and pen marks as two separate problems. Hair is detected by its shape: it is dark and elongated, so we use morphological filtering. Pen marks are more variable in shape but almost always a specific colour (blue, purple, or black ink) so we detect them using HSV colour segmentation. This means we can remove pen marks reliably without accidentally removing dark hair, which a shape-only approach could struggle to do.

We also investigate an open question of our own choosing: *Can texture features improve lesion classification?* Some cancerous lesions have a rough or scaly surface texture that does not show up in colour or shape measurements. We test this with a third model that adds Local Binary Pattern (LBP) texture features on top of the extended set, and keep it separate so we can see exactly what difference it makes.

1.1 Contributions

Our main contributions in this report are:

- A feature extraction pipeline covering asymmetry, border, colour (including blue-white veil score and melanoma colour count), and artefact coverage.
- Separate detection and removal pipelines for hair and pen marks, using methods suited to each artefact type.
- A manual annotation study on hair and pen mark presence to validate our detection methods, with Fleiss’ Kappa inter-rater analysis.
- A three-model comparison across two classifiers to answer both the research question and the open question.

2 Related Work

The most successful skin lesion classifiers today use convolutional neural networks(CNN),

which can outperform dermatologists on dermoscopy benchmarks (Shetty et al., 2022). We chose not to go in that direction. These models require large amounts of labeled data, and their internal reasoning is hard to inspect, which makes it difficult to understand why a prediction was made. Instead we build on the classical feature-based approach, where every prediction can be traced back to a specific measurable property of the image.

The features we use are based on how dermatologists actually think about lesions. The ABCD rule (Nachbar et al., 1994) breaks down suspicious signs into four criteria, and Kasmi and Mokrani (Kasmi & Mokrani, 2016) show that these can be turned into reliable computable features. For colour, Celebi et al. (Celebi et al., 2007) demonstrate that HSV statistics, especially saturation variance, are among the strongest signals for melanoma. We use both of these as a starting point, but adapt the colour features in two ways. First, rather than matching pixels against a fixed palette, we check which of the six canonical melanoma-associated colours defined by Kasmi and Mokrani (Kasmi & Mokrani, 2016) are present in the lesion and count them. These six colours (white, black, red, light brown, dark brown, and blue-grey) are clinically established indicators of malignancy, and counting how many appear gives a score from 0 to 6 that varies meaningfully between lesions. Second, we include a blue-white veil score, a recognized dermoscopy sign of melanoma that can be approximated directly from RGB channel differences (Argenziano et al., 2003).

Previous studies have shown that artifacts such as hair and surgical pen markings can bias skin lesion classifiers by introducing patterns unrelated to the lesion itself. (Winkler et al., 2019), for example, showed that a classifier trained on images with surgical pen markings learned to rely on the marks themselves rather than the lesion, and its performance collapsed when the marks were removed. Hair removal has been studied since at least Lee

et al. (Lee et al., 1997), who established morphological filtering as a common approach for hair removal. However, hair and pen marks differ visually: hair is mainly characterized by elongated structural patterns, whereas pen marks are more strongly defined by their color properties. That is why we treat these artifacts separately instead of applying a single preprocessing method to both.

3 Data

3.1 PAD-UFES-20

We use the PAD-UFES-20 dataset (Pacheco et al., 2020), which contains 2,298 images from 1,373 patients collected at a dermatology clinic in Espírito Santo, Brazil. Each image is labelled with one of six lesion types: Actinic Keratosis (ACK), Basal Cell Carcinoma (BCC), Malignant Melanoma (MEL), Melanocytic Nevus (NEV), Squamous Cell Carcinoma (SCC), and Seborrheic Keratosis (SEK). MEL only has 52 samples out of 2,298, which would make a six-class model heavily dominated by the larger categories and the model would barely learn anything useful about the rarest class. Therefore, we merged the six types into two groups, which reflects the clinical goal of separating dangerous lesions from benign ones:

- **Cancerous:** BCC, SCC, MEL
- **Non-cancerous:** ACK, NEV, SEK

3.2 Dataset Exploration

Table 1 shows the resulting class counts. The split is slightly imbalanced, with 1,089 cancerous and 1,209 non-cancerous cases. We address this through class-weight balancing in the classifiers.

Table 1: Class distribution in PAD-UFES-20 after binary grouping. Cancerous cases make up roughly 47% of the total.

Class	Lesion types	Count
Cancerous	BCC, SCC, MEL	1,089
Non-cancerous	ACK, NEV, SEK	1,209
Total		2,298

Images without a segmentation mask are skipped, since our features are all computed inside the lesion region. Going through the data we noticed that hair and pen marks are present in a significant number of images, often overlapping the lesion itself. This is what originally motivated the research question. We also noticed that cancerous and non-cancerous lesions can look very similar in terms of shape alone. This is consistent with observations in prior work (Kasmi & Mokrani, 2016) and is why color plays a central role in our feature set.

3.3 Annotation of Hair and Pen Marks

To check whether our automatic detection was picking up what a human would actually call hair or pen marks, we use an annotation study conducted collectively across all project groups in the course. Each image was independently annotated by up to five human evaluators, and the results were merged into a single shared dataset. In total, 1,637 of the 2,298 images were annotated, as each group labeled a subset of the full dataset. The labels used were:

- **0** – no or very little hair/pen marks
- **1** – some hair/pen marks
- **2** – a lot of hair/pen marks

We measured agreement using Fleiss’ Kappa (Landis & Koch, 1977), which is designed for cases with more than two raters and corrects for chance agreement. A kappa of 0 means raters agreed no better than chance, while 1 means perfect agreement. We got $\kappa = 0.687$, corresponding to substantial agreement. The

computation is reproducible from the repository using `compute_kappa.py`. The main source of disagreement was between labels 0 and 1, since the line between “barely any” and “some” is genuinely ambiguous, and different people draw it differently. There were no conflicts between the extremes (0 vs. 2), which means the labels are most reliable for telling heavy artifacts presence apart from little or none. We use the majority-vote annotations to evaluate whether the automatically computed coverage features correspond to human judgement covered in Section 4.4.

4 Feature Extraction

All features are computed inside the lesion region defined by the segmentation mask, so background pixels should not affect any of our measurements. We go through each feature group below, covering what we compute, why we chose it and what the distributions looks like.

4.1 Asymmetry

The A in ABCD stands for asymmetry. The clinical idea is that malignant lesions tend to look different on one side than the other, while benign ones are more symmetric (Nachbar et al., 1994).

To calculate asymmetry, we first find the bounds of the lesion in the mask and locate the geometric centre of that bounding box. We then crop a square window around the centre, sized dynamically to fit the lesion with a small margin. This crop is then compared against its horizontal mirror image using the following formula:

$$\text{Asymmetry} = \frac{\sum |M - \text{flip}(M)|}{\max(\sum M, 1)} \quad (1)$$

where M is the binary mask crop and $\text{flip}(M)$ is its left-right reflection. The result is capped at 1 and rounded to four decimal places. A score of 0 means perfect symmetry; higher values mean more asymmetry.

In practice, asymmetry alone is not enough to determine a diagnosis, so it works best as part of a combined feature set alongside colour.

4.2 Border

We extract five shape descriptors from the largest outline in the binary mask, found using OpenCV:

- **Area** and **perimeter** of the lesion outline.
- **Compactness:** $\frac{\text{Perimeter}^2}{4\pi \cdot \text{Area}}$. A perfect circle gives a value of 1; higher values mean more irregular shapes. Malignant lesions tend to have irregular borders and therefore higher compactness (Kasmi & Mokrani, 2016).
- **Solidity:** area divided by convex hull area. A value near 1 means the lesion fills its hull; lower values indicate concavities or jagged edges.
- **Eccentricity:** derived from the fitted ellipse. Near 0 means circular; near 1 means highly elongated.

Like asymmetry, border features alone do not cleanly separate the two classes. This is a known challenge with clinical images (Kasmi & Mokrani, 2016), and is why colour features play a central role in our pipeline.

4.3 Colour

Prior work has shown that colour is particularly informative for distinguishing cancerous from non-cancerous lesions (Celebi et al., 2007). We extract 15 colour features from the lesion pixels:

- **Mean and std of R, G, B** (6 features): basic per-channel statistics. High standard deviation means the colour varies a lot across the lesion, which tends to be more common in malignant cases.

- **Mean and std of H, S, V** (6 features): the same statistics in HSV colour space. HSV separates colour information from brightness so these features are less affected by lighting differences between images. With higher saturation variance, there is said to be higher chance of malignancy (Celebi et al., 2007).

- **Colour entropy** (1 feature): computed from a 32-bin greyscale histogram inside the lesion mask. High entropy means the intensity varies a lot across the lesion.

- **Blue-white veil score** (1 feature): the fraction of lesion pixels where the blue channel value is higher than the red. Blue-white veil is a recognised dermoscopy sign of melanoma, caused by melanin deposits in the dermis (Argenziano et al., 2003). We approximate it directly from the RGB values.

- **Melanoma colour count** (1 feature): we check which of the six canonical melanoma-associated colours are present in the lesion and count them. These colours (white, black, red, light brown, dark brown, and blue-grey) are clinically established indicators of malignancy (Kasmi & Mokrani, 2016). The result is a score from 0 to 6.

Mean hue shows the clearest separation between classes, with cancerous lesions tending towards higher hue values. Saturation standard deviation also differs, with cancerous lesions showing more spread, consistent with prior work on colour mixing (Celebi et al., 2007). Colour entropy and blue-white veil score overlap considerably between classes. Figure 1 shows the distributions for these four features.

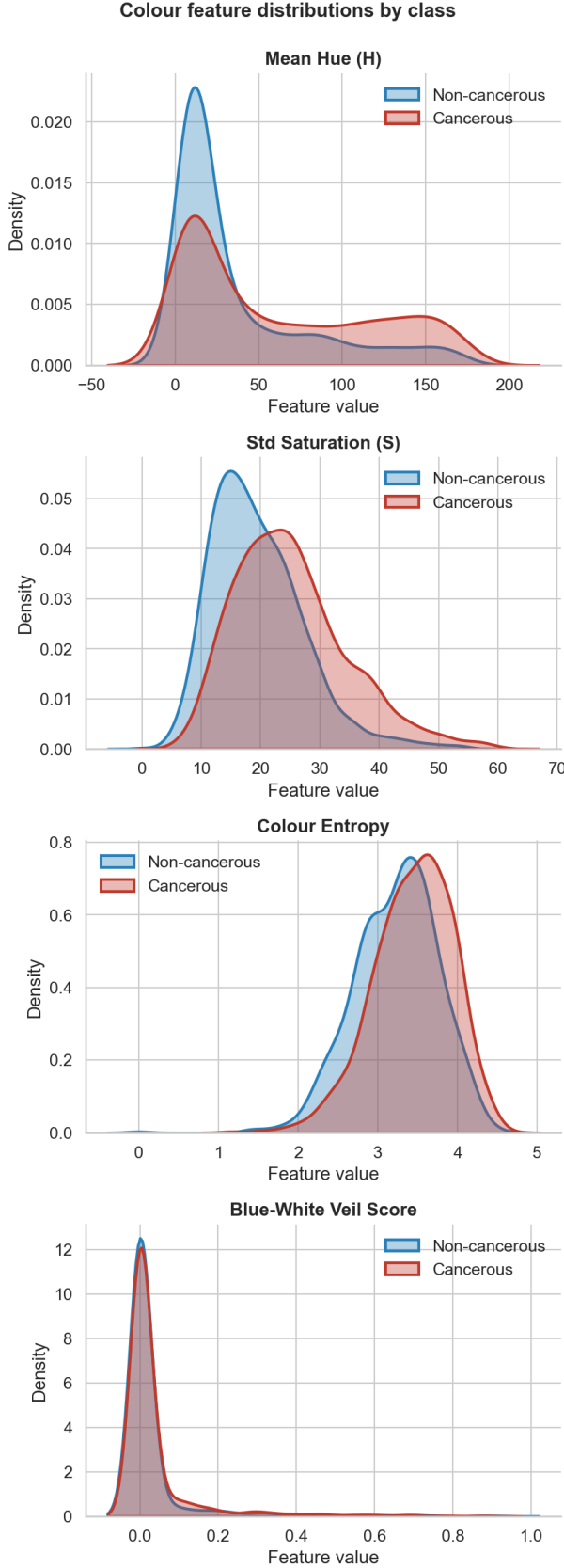


Figure 1: Selected colour feature distributions by class (cancerous in red, non-cancerous in blue). Mean hue and saturation standard deviation show the most visible separation between classes.

4.4 Hair and Pen Mark Coverage

We compute a coverage value for all images, that says how much of the image is covered by each type of mark. These serve two purposes: as input features in the extended model, and as thresholds that control how aggressively (if at all) the removal step is applied.

Hair coverage is computed from the greyscale image using a black-hat morphological transform with a 5×5 cross-shaped kernel. Black-hat finds dark elongated structures against a lighter background, which is a reasonable description of dark hair on skin. Then we set the threshold at 10 to create a binary hair mask, and the coverage value is the fraction of image pixels that fall inside that mask.

Pen mark coverage uses a different method. We convert the image to HSV and look for two colour ranges: blue and purple hues (OpenCV hue 100 to 160, covering most clinical pen colours) and very dark values (value channel below 60 to catch black pens). The two masks are combined and then cleaned by eliminating small noise pixels while keeping the actual pen marks intact. The reason we use colour rather than shape here is that pen marks and dark hair can look very similar in terms of shape, but they can be distinguished by their different colours, which lets us separate them cleanly.

Figure 2 shows hair coverage values against the manual annotation labels from Section 3.3. Images annotated as heavy hair (label 2) show a wider spread of values that lean more towards higher coverage values. Labels 0 and 1 overlap heavily at low coverage values, which is consistent with the annotation ambiguity we noted earlier. This confirms that the coverage feature is most reliable for detecting heavy hair presence rather than telling apart "little" and "some hair" cases.

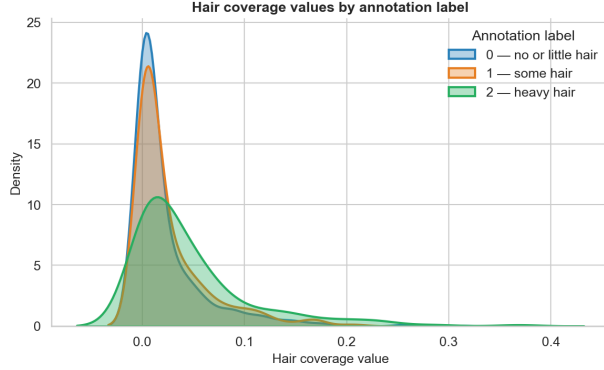


Figure 2: Hair coverage values by manual annotation label (0 = no or little hair, 1 = some hair, 2 = heavy hair). Label 2 is shifted towards higher coverage values, while labels 0 and 1 overlap at low values, reflecting the annotation ambiguity noted in Section 3.3.

5 Classification

5.1 Model Definitions

We define three feature sets corresponding to our three models. The difference between baseline and extended is the preprocessing step and the addition of two coverage features, which allows us to directly answer the research question.

- **Baseline model:** asymmetry, area, perimeter, compactness, solidity, eccentricity, mean and std of RGB and HSV (12 features), colour entropy, blue-white veil score, and melanoma colour count. The features are extracted from the *original* images without preprocessing.
- **Extended model:** the same 17 features, extracted from images where pen marks and hair have been removed beforehand. Hair coverage and pen mark coverage are added as two extra features, giving 19 in total. This model directly tests the effect of hair and pen mark removal.
- **Texture model (open question):** all 19 extended features plus 16 LBP histogram bins, giving 35 features in total. We keep this as a separate third model

so the contribution of texture can be assessed independently.

5.2 Preprocessing: Hair and Pen Mark Removal

For the extended and texture models, we apply hair and pen mark removal steps before feature extraction. Pen marks are removed first because they tend to be larger and that gives us a cleaner starting point for the hair step.

Pen mark removal uses the same HSV colour segmentation as the coverage detection. The mask is slightly expanded before inpainting so that the edges of the marks are also covered. The removed pixels are filled in using TELEA fast marching inpainting (Telea, 2004), which fills the gaps by spreading the texture from the surrounding skin inward.

Hair removal uses black-hat filtering on the greyscale image. The kernel size and threshold are adapted based on the hair coverage value so that heavily affected images get a more aggressive filter, while lightly affected ones are processed more gently:

- Coverage below 0.02: kernel 5×5 , threshold 15, inpaint radius 3.
- Coverage 0.02 to 0.06: kernel 7×7 , threshold 10, inpaint radius 5.
- Coverage above 0.06: kernel 11×11 , threshold 8, inpaint radius 7.

Hair removal is skipped entirely when the coverage is below 0.01, and pen mark removal is skipped below 0.005. This avoids unnecessarily processing clean images. Figure 3 shows the pipeline on a few example images.

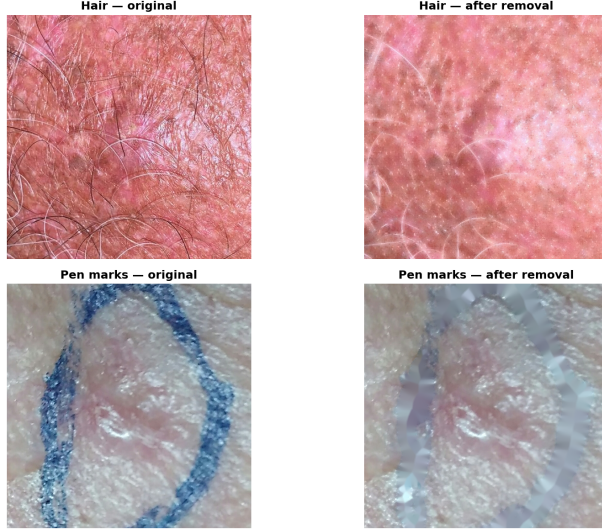


Figure 3: Preprocessing pipeline examples. Top row: image with dark hair, before and after removal. Bottom row: image with pen marks, before and after removal.

5.3 Classifier Selection

We compare three classifiers across all three feature sets.

Random Forest (RF) builds many decision trees, each of which is trained on a random subset of the data and a random set of features. It handles complex and non-linear relationships between features well and is stable even with a mixed dataset like ours, with different scales and distributions. We use 200 trees, set a maximum depth of each tree to 10 and use class-weight balancing so the model does not favor the larger group. We initially expected RF to perform better, but cross-validation showed LR was stronger across all three feature sets, suggesting the features contain patterns that are simple enough for a linear classifier to handle.

Logistic Regression (LR) is a linear classifier that we include for comparison. Its coefficients directly show which features push the prediction towards cancer and which push it away, making it easier to interpret. We use L_2 regularisation with $C = 1.0$ and class-weight balancing. If it performs reasonably well, it suggests the features contain signal that is at least partially linearly separable.

K-Nearest-Neighbor (KNN) works by identifying the K closest data points to a given input, in the feature space, and classifies the input by the average values or majority class of those data points. We chose a k -value of $K = 7$. This means that for our input (the features of a lesion), the model identifies the 7 closest data points from the training data, and classifies the lesion from the majority class from those data points.

5.4 Experimental Setup

Data splitting. We split the data 80/20 into a development set and a held-out test set using `GroupShuffleSplit` with patient ID as the grouping variable. This makes sure all images from the same patient are in the same split. This matters because PAD-UFES-20 contains multiple images per patient, and a standard random split could put images of the same lesion in both training and test, which would give inflated results that do not reflect real generalisation to new patients.

Cross-validation. We compare classifiers using 5-fold `GroupKFold` cross-validation on the development set, again keeping all images from the same patient within a single fold. We report mean and standard deviation of AUC across folds.

Evaluation metrics. We use AUC as our primary metric and accuracy as a secondary one. AUC measures how well the model ranks cancerous lesions above non-cancerous ones at all possible thresholds, rather than at one fixed cut-off. This is more useful in a medical context because the appropriate threshold depends on the intended use: a screening tool might accept more false positives to avoid missing any cancers. AUC is also less sensitive to class imbalance than accuracy alone (Fawcett, 2006).

Standardisation. All features are standardised to zero mean and unit variance. The scaler is fitted on the training data only and then applied to the validation or test set, so there is no information leaking across splits. Missing values which can be caused when a

mask produces an invalid outline, are replaced with the training set mean.

6 Evaluation and Results

6.1 Cross-Validation Results

Table 2 shows the 5-fold cross-validation results on the development set for all three classifiers across all three feature sets.

Table 2: 5-fold cross-validation results on the development set. Mean $\pm$ std AUC and accuracy across folds. Best AUC per feature set in **bold**.

Feature set	Classifier	AUC	ACC
Baseline	Random Forest	0.780 ± 0.034	0.701
	Logistic Regression	0.795 ± 0.028	0.718
	KNN ($k = 7$)	0.708 ± 0.032	0.657
Extended	Random Forest	0.776 ± 0.030	0.704
	Logistic Regression	0.796 ± 0.027	0.723
	KNN ($k = 7$)	0.712 ± 0.031	0.658
Texture	Random Forest	0.784 ± 0.029	0.710
	Logistic Regression	0.806 ± 0.023	0.730
	KNN ($k = 7$)	0.723 ± 0.022	0.662

Logistic Regression was the best classifier across all three feature sets, which was the opposite of what we expected going in. KNN performed worst in all cases. The gap between LR and RF was small but consistent, and LR also showed lower standard deviation across folds, indicating more stable performance. Based on this, we selected Logistic Regression for the final test set evaluation.

6.2 Test Set Results

Based on cross-validation, we selected Logistic Regression as the best classifier and re-trained it on the full development set. Table 3 reports the final test set results for all three models.

Table 3: Test set results for Logistic Regression per feature set. AUC is our primary metric. Accuracy is shown for completeness.

Feature set	AUC	ACC
Baseline	0.828	0.766
Extended	0.824	0.758
Texture	0.813	0.756

6.3 Effect of Hair and Pen Mark Removal

The difference in AUC between the baseline and extended models answers our research question directly. The baseline model achieves an AUC of 0.828 compared to 0.824 for the extended model. The difference is very small and within the observed variation (see Table 2), meaning that the decrease is unlikely to be statistically significant and that the preprocessing does not change model performance on this data set significantly. Therefore removing hair and pen marks before feature extraction did not improve performance. One explanation is that inpainting creates its own artificial patterns. First, it removes hair and pen marks and then replaces them with patterns that might not necessarily match the true skin structure, which can distort colour and texture features. It is also possible that the classifiers are already immune to the distortions present in this dataset, since features are averaged over the whole lesion, so small marks may not shift aggregate statistics such as mean colour or entropy much. Furthermore, hair or pen marks might actually correlate with the target variable. For example, clinicians may be more likely to mark suspicious lesions, which could introduce a weak correlation between pen marks and malignancy.

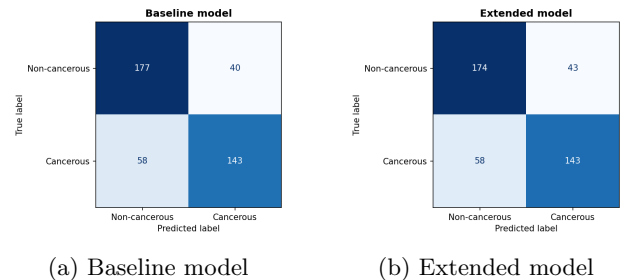


Figure 4: Confusion matrices on the test set. Rows are true labels, columns are predicted labels. Both models output the same number of false negatives (missed cancers): 58

Figure 4 shows the confusion matrices for the baseline and the extended model. Both models output the same number of false nega-

tives (58 cases), indicating that preprocessing does not improve the model’s ability to detect cancerous lesions. This is important because reducing false negatives is a key priority in medical diagnosis.

Overall, these results suggest that hair and pen marks removal is not always beneficial, and that its effectiveness depends on the balance between noise reduction and distortion introduced by preprocessing. In this case, the removal process does not improve classification performance, highlighting the importance of evaluating preprocessing steps through actual testing rather than assuming that they will be helpful.

7 Open Question

Can texture features improve lesion classification?

When going through the images we noticed that some cancerous lesions have a rough or scaly surface texture, which does not show up in colour statistics or shape measurements in a clear way. Colour tells you what the lesion looks like on average, and the shape describes its outline. Neither captures the structure of the surface itself. We wanted to test whether a texture descriptor could add something useful on top of the features we already have.

We use Local Binary Patterns (LBP) (Ojala et al., 2002) to learn about the texture. The idea is simple: for each pixel in the image, you compare it to its surrounding neighbors and write down which ones are brighter and which are darker. This gives a binary code per pixel that encodes the local texture pattern. We use the uniform variant, which groups similar rotation patterns together and makes the features rotation-invariant. This is important since lesions can appear in any orientation. We set the radius to 3 and use 24 comparison points, then bin all the codes from pixels inside the lesion mask into a 16-bin normalized histogram. This gives us 16 extra features per image.

We keep it as a separate third model so the

comparison against the extended model directly shows what texture adds, without anything else changing between the two.

Table 3 in Section 6 shows the results.

The texture model achieved an AUC of 0.813, slightly below both the baseline (0.828) and the extended model (0.824), which means adding LBP features did not improve performance. LBP operates at a very local pixel scale, which can be noisy in these images. It is also possible that colour already captures most of the relevant information in this dataset. Future work could explore texture descriptors that work at larger spatial scales, such as Gabor filters.

8 Discussion and Conclusion

8.1 Summary

Removing hair and pen marks before feature extraction did not improve model performance. The baseline model achieved an AUC of 0.828 compared to 0.824 for the extended model. Adding LBP texture features did also not help, with the texture model scoring 0.813. Logistic Regression outperformed Random Forest across all three feature sets, which was unexpected and suggests the decision boundary between cancerous and non-cancerous lesions is more linear than we originally thought. Overall, colour features seem to carry most of the useful information in this dataset.

8.2 Limitations

There are a few things that limited what we could conclude from this study.

The segmentation masks vary in quality. Some do not follow the lesion boundary accurately, which adds noise to all features regardless of how well the preprocessing runs. There is not much we can do about this without remaking the masks manually.

Our hair removal only works for dark hair. Images with light hair or a mix of light and dark hair are not handled, which means the

extended model benefits less for those images. This likely reduces the level of improvement that we see overall.

The pen mark detector targets blue, purple, and black ink. Faded marks or unusual colours would not be caught. We also noticed that the dark colour range can occasionally pick up very dark lesion pixels not just marks, which could slightly distort colour features in those cases.

The annotation study revealed that the boundary between labels 0 and 1 is subjective and hard to define consistently. This limits how well we can validate the coverage features for images with only a little hair or a few faint marks.

The dataset comes from one clinic in Brazil. Whether our pipeline generalises to other clinical settings, devices, or patient populations is unknown.

8.3 Future Improvements

The clearest improvement to the hair removal would be adding a top-hat transform alongside the black-hat, which would detect both light and dark hair (Lee et al., 1997). This would extend coverage to more of the dataset without changing the overall approach.

Better segmentation masks would help every part of the pipeline. Even a semi-automatic correction tool would reduce the noise coming from incorrect mask boundaries.

For texture, LBP works at a very local pixel scale. Gabor filters or Histogram of Oriented Gradients operate at larger spatial scales and might be more indifferent to image noise (Dalal & Triggs, 2005). It would also be interesting to include patient meta-data such as age, Fitzpatrick skin type, or le-

sion diameter, since this is already available in the dataset and could complement the image-based features.

8.4 Broader Perspective

Tools like ours could genuinely be useful in settings where access to dermatologists is limited. But this project is also a reminder that good performance on a training dataset does not automatically mean something that works in the real world. A classifier could learn from the presence of pen marks or hair patterns rather than from the lesion itself and if that is happening, performance would drop at any clinic that handles images differently (Winkler et al., 2019). Bissoto et al. (Bissoto et al., 2019) show this is a common problem in medical imaging more broadly. Addressing it explicitly, as we try to do with our pre-processing is part of what makes a pipeline trustworthy rather than just accurate on one dataset.

8.5 Conclusion

We set out to answer one question: does removing hair and pen marks before extracting features help a skin lesion classifier? We built three models to find out: a baseline on the original images, an extended model on cleaned images, and a texture model that adds LBP features. We evaluated all three using patient-aware cross-validation to avoid inflated results. Our results show that pre-processing did not improve performance, and that texture features did not add value on top of that. The most useful information appears to come from colour features and a simple linear classifier was sufficient to capture it.

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